Regression assignment =

Stage 1: domain: Machine Learning
Stage 2: supervised Learning
Stage 3: Regression

the client has given the dataset where the insurance charges varies based on age , sex, bmi, children, smoker and the charges varies accordingly.
and the dataset has 1339 rows and 6 columns

preprocessing: two of the columns have categorical data like sex and smoker and they are nominal data so we need to use one hot encoding Technique and dataset doesn't have big numbers so no need to preprocess the rest of the columns

Dataset filename: **insurance_pre.csv**
R2_score comparison for different algorithms like Multiple Linear Algorithm, Support Vector Machine, Decision Tree and Random Forest by creating multiple models with different hyper parameters are below:

1) Multiple Linear Algorithm:

R2value is 0.7894790349

2) SVM (Regression) Algorithm:

S. No	Hyper parameter	LINEAR (r value)	RBF (NON LINEAR) (r value	POLY (r value)	SIGMOID (r value)
1	C10	0.4624684	-0.0322732	0.03871	0.039307
2	C100	0.6288792	0.3200317	0.617956	0.527610
3	C500	0.7631057	0.6642984	0.826368	0.444606
4	C1000	0.7649311	0.8102064	0.856648	0.287470
5	C3000	0.7414230	0.8663393	0.859893	-2.124419
6	C15000	0.7414230	0.8775396	0.857965	-62.8488

the SVM regression best R2value in (Linear and hyper parameter C=15000) is = 0.8775396

3) Decision Tree Algorithm:

S.NO	CRITERION	MAX FEATURES	SPLITER	R VALUE
1	squared_error	sqrt	best	0.5980887
2	squared_error	sqrt	random	0.6913106
3	squared_error	Log2	best	0.6773391
4	squared_error	Log2	random	0.7280811
5	squared_error	none	best	0.6845671
6	squared_error	none	random	0.7072994
7	friedman_mse	sqrt	best	0.6959088
8	friedman_mse	sqrt	random	0.7309781
9	friedman_mse	Log2	best	0.7085692

10	friedman_mse	Log2	random	0.6315428
11	friedman_mse	none	best	0.7003293
12	friedman_mse	none	random	0.6649652
13	absolute_error	sqrt	best	0.6929682
14	absolute_error	sqrt	random	0.622836
15	absolute_error	Log2	best	0.7245908
16	absolute_error	Log2	random	0.6519101
17	absolute_error	none	best	0.6903801
18	absolute_error	none	random	0.6894519

the Decision Tree best R2value in (criterion= friedman_mse, Max features= sqrt, splitter=random) is = 0.7309781.

4) Random Forest Algorithm:

S.NO	CRITERION	MAX FEATURES	n_estimators	R VALUE
1	squared_error	sqrt	10	0.8520006
2	squared_error	sqrt	100	0.8710271
3	squared_error	Log2	10	0.8520006
4	squared_error	Log2	100	0.8710271
5	squared_error	none	10	0.8330304
6	squared_error	none	100	0.8538307
7	friedman_mse	sqrt	10	0.8502777
8	friedman_mse	sqrt	100	0.8710544
9	friedman_mse	Log2	10	0.8502774
10	friedman_mse	Log2	100	0.8710544
11	friedman_mse	none	10	0.8331662
12	friedman_mse	none	100	0.8540518
13	absolute_error	sqrt	10	0.8574290
14	absolute_error	sqrt	100	0.8710681
15	absolute_error	Log2	10	0.857429
16	absolute_error	Log2	100	0.8710685
17	absolute_error	none	10	0.8350635
18	absolute_error	none	100	0.8520093

the Random Forest best R2value in (criterion= absolute_error, Max features= log2, n_estimators = 100) is = 0.8710685.

SVM (Regression) Algorithm gives the best R2 score =0.8775396, so it the best model